

Exploring the Relation between Interest Rate and Stock Market Indices – An ARDL Approach

Jitendra Kumar Dixit¹, Dr. Sanjeev Gupta²

¹ Assistant Professor, Institute of Business Management, GLA University, Mathura(U.P.), India

² Professor, MLB Govt. College of Excellence, Gwalior (M.P.), India

E mail: jitendra.dxt@gmail.com

Abstract:

Stock market tells the story of growth of any economy around the globe. Apart from numerous economic factors stock market movement are controlled and directed by interest rate change. That attracts the researchers around the globe to explore the relation between the interest rate and stock market movement. Results of the previous studies ranges between positive and negative even few studies supported no relation between interest rate and stock market movement. These variations in the relation motivate to re-investigate the relation between both the variables.

For the analysis, daily data of both interest rate and closing value of stock market indices (Sensex), ranging from April 2011 to March 2016 is used. Database of interest rate and stock market indices is collected from the web portal of Reserve Bank of India (RBI) and Bombay Stock Exchange (BSE). For the analysis, unit root test followed by ARDL bound test is used to examine the co-integration between interest rate and stock market indices. This paper uses Vector auto regressive model (VAR) to characterize the dynamics of stock market indices to interest rate shocks.

Key Words- VAR, interest rate, inflation, Sensex

Jitendra Kumar Dixit¹, Dr. Sanjeev Gupta²

¹ Assistant Professor, Institute of Business Management, GLA University, Mathura (U.P.), India

² Professor, MLB Govt. College of Excellence, Gwalior (M.P.), India

E mail: jitendra.dxt@gmail.com

1. Introduction

Stock market of any country tells the story of growth of that economy around the globe. Efficiency of any market can be measured and judged by the way market accommodate and adjust the economic shocks. Stock market is presumed as a barometer to ensure the economic health and confidence for the global and domestic investors (Alam and Uddin, 2009). Information availability and accessibility makes the investors more dynamic and predict the efficiency of the market (Maysami et. Al., 2004).

From the last two decades, several research and theory advocated that the stock market movements are highly sensitive towards macro-economic policies. But these studies were limited up to the developed market i.e. Fama (1981) and Chen (1991) for the US market, Hamao (1988) on Japanese market and Poon and Taylor (1992) on the UK market. Impact of monetary policy announcement on stock market performance (Pearce and Roley, 1983; Hardouvelis, 1987; McQueen and Roley, 1993; Rigobon and Sack, 2002; Bernanke and Kuttner, 2005, inter alia) is empirically supported by several researches but theoretically there is ambiguity whether the cash flow effect or the discount rate effect dominates.

Monetary policy framework was initiated in India as interim liquidity adjustment facility (ILAF) in April 1999 which transitioned in to full-fledge liquidity adjustment (LAF) in June 2000, with the purpose of managing capital flow. Repo rate and reverse repo emerged as a key interest rate (short-term) and primary instrument for interest rate signaling. Repo rate, a short-term interest rate is a macroeconomic variable that is used by the monetary authorities for achieving and maintaining equilibrium within the economy. Smooth implementation and execution of monetary policy ensure the overall economic growth. Stock market indices are integrated with macroeconomic variables and sensitivity of the market is reflected with the changes in expectation about future prospects (Ahmed, 2008). Discount rate changes in

developed market as a proxy of monetary policy stance grab a widespread attention in guiding and controlling the decision of financial institutions in fixing their interest rate and have impact on the stock market returns. Frequent changes provide a signal to market and restrictive or expansive direction to monetary policy. Central bank established the discount rate as per the liquidity and credit need of the economy. Rise in interest rate is due to discount rate which in-turns results decline in prices of stocks and vice-versa (Waud, 1970; Jensen et. Al.1996).

Loyaza and Schmidt-Hebbel (2002) supported the relevancy of traditional interest rate channel in influencing output and price and recent survey by Boivin et.al. (2010) also supported the neo-classical channel (direct interest rate) as the core channel in macro-economic modeling.

Different proxies of the short-term interest rate other than the repo rate were used in most of the studies that influences the output and witness the ambiguity in the results. This study centrally focuses around investigating dynamic nexus of interest rate and stock market performance in Indian perspective. The organization of the study is provided in Section 1, literature review is carried out in Section 2. Hypothesis and methodological framework is explained in Section 3. The estimation and interpretation of results is mentioned in Section 4, Results and discussion in Section 5, and section 6 concludes the paper.

2. Review of Literature

The contribution of well functioning stock market contribution on economic growth and development can never be ignored. According to Malkiel (1998), higher stock value create wealth, reduces the cost of raising the fresh fund that provides an incentive to firms in increasing investment spending and booming market increases investors confidence and fostering spending that in turn contribute in achieving higher economic growth. Levine (1997) in their study also claims the importance of the developed market as they provide easy and quick liquidity that makes the investment less risky and more attractive. Stock market performance depicts the true picture about the growth of any economy. The Stock market performance around the globe are fully guided and controlled by the interest rate as the financial instrument and market is highly sensitive towards the interest rate movement. The impact assessment of interest rate on stock market attracts the researcher to explore the facts that have ambiguity varying from market to market and country to country.

Alam and Uddin (2009) conducted a study on developed and developing economies to explore the impact of interest rate on stock prices and result concluded negative relationship between both the variables that if the interest rate will rise (low), market will move downward (upward). Similar results are supported by Moya-Martínez, et al. (2003) in their study on Spanish stock market and Uddin and Alam (2007) during their study on Dhaka Stock Exchange (DSE). Kim (2003) during their study on US S& P 500 index reported negative association with interest rate. Similar findings are supported by Rigobonet at.(2004) that rise in short-term interest rate will lead to decline in stock price and bond yield curve. Investor's decision related to the investment in bonds or stock is controlled and guided by interest rate changes (Apergis and Eleftheriou, 2002). Investors prefer bond as price of stock decreases with higher interest rate. On other side, increase in the stock prices caused reduction in interest rate. This negative relationship is supported by Gjerde and Saettem (1999), Paul and Mallik (2003), Wongbangpo and Sharma (2002), Nasseh and Strauss (2004), Puah and Jayaraman (2007), McMillan (2005) and Reilly et al. (2007). Fama (1981) in their study confirm the negative relation between stock market returns and short-term interest rate which is proxied by inflation. Arango (2002) find evidences of inverse relationship between stock market and inter bank loan interest rate and the same results are supported by Hsing (2004).

Many studies related to different economy have documented that financial market volatility are the reflection of interest rate changes. Flannery and James (1984) explored the relation between stock market return to interest rate and confirmed the negative relationship between both whether short run or long run. Lee, Yong and Zhang (2006) also confer negative association ship in long term intervening interest rate and stock prices. On the contrary, Liu and Shrestha (2008) in their study supported no relationship means interest rate do not hit on stock market in china. Pilinkus and Boguslankas (2009) analyze the relationship and concluded negative behavior of interest rate on stock prices. Humpe and Macmillan (2007) advocated contradictory long term association enclosed by stock prices and interest rate in US. This relationship is also supported by Maysami, Howe and Hamzah, (2004) between stock returns and long term interest rate in Singapore.

But Ratanapakorn and Sharma (2007) in their study supported positive association intervening short-term interest rate and stock prices and negative relationship between stock prices and interest rate in US. Moya-Martínez, et al. (2003) makes a study on Spanish stock

market and reported negative interest rate sensitivity with stock returns. A positive relationship was also supported by Lobo (2002), Apergis and Eleftheriou (2002), Erdem et al. (2005) and Bohl et al. (2007). Lobo (2002) suggested FED's disclosure policy as a leading factor of stock market volatility. Rise in interest rate by FED than expected, adversely affect the stock market. Similar findings are suggested by (Erdem et al., 2005) in their study on Istanbul Stock Exchange. Bohl et al. (2007) concluded that the positive association relies on the heteroskedasticity in stock returns and interest rates.

To address the issue of stock market and interest rate, Panda (2008) analyzed the relation and examine the direction of causality between both the variables in Indian stock market. The result advocated the long-run association between both the variables. A unidirectional long-run causality is observed from interest rate to stock price whereas from long-term interest rates to stock prices, short-run causality. Harasty and Roulet (2000) in their study on 17 developed countries concluded that stock price and long-term interest rate are co-integrated. Rigobon and Sack (2003) empirically confers that monetary policy significantly reacts to stock market movements. An increase in short-term interest rate results decline in stock prices and stock market performances are significantly influenced by monetary policy changes and guide market movements. Lee (1997) in their empirical work explored that the relation between short-term interest rate and stock market is not stable over time. It gradually changes from a significantly negative to positive although insignificant relationship or even no relationship.

This variation in the results of different studies motivates to re-investigate the relation between interest rate and stock market movement.

3. Methodology-

Repo rate, proxy of interest rate is used as a dependent variable and closing value of stock market indices (BSE Sensex) is treated as independent variable. The daily data related to interest rate is collected from the web portal of RBI (Reserve Bank of India) and daily data of closing value of stock market indices are collected from the web portal of BSE. To address the issue, this study includes daily, weekly, monthly and quarterly data of interest rate and natural log of the stock market indices from April 2011 to March 2016.

3.1. Hypothesis

In the literature the relation between interest rate and stock market is oscillating between positive and negative. Even Liu and Shrestha (2008) in their study supported no relation between both the variables. These differences in the results across various studies might be evident because of the type and the nature of data (daily, weekly, monthly, and quarterly, annually) considered during the study. To explore the fact daily data of both the variables are converted in to weekly, monthly and quarterly time series and are analyzed to re-investigate the relationship. Based on the literature following hypothesis are framed:

H₀₁: Interest rate and Stock market are co-integrated.

H₀₂: Interest rate and stock market are negatively correlated.

H₀₃: Interest rate granger causes stock market.

H₀₄: The relation between interest rate and stock market are independent of the nature of data.

4. Method of Data Analysis

To check the stationary properties (unit roots) of time series ADF test (Augmented Dickey-Fuller, 1979) is employed. First difference regression of the series can expressed as-

$$\Delta Z_t = \alpha_1 Z_{t-1} + \sum_{j=1}^p \beta_j \Delta Z_{t-j} + \varepsilon_t \quad \dots\dots\dots (1)$$

Hypothesis of a unit root is rejected if the coefficient is something different from zero. Null hypothesis rejection reflects stationary.

The ARDL bounds testing procedure and VECM approaches are used for testing the hypothesis. The ARDL bounds testing approach is used to examine the long-run co integration relationship between the variables. The ARDL approach is involved to estimate the following unrestricted VECM by assuming that Z and W are two variables:

$$\Delta Z_t = \lambda_0 + \sum_{i=1}^p \theta_i \Delta Z_t + \sum_{i=0}^p \lambda_i W_{t-i} + \alpha_1 Z_{t-1} + \alpha_2 W_{t-i} + v_1 t \quad \dots\dots\dots (2)$$

$$\Delta W_t = \gamma_0 + \sum_{i=1}^p \delta_i \Delta W_{t-i} + \sum_{i=0}^p \gamma_i \Delta Z_{t-i} + \beta_1 Z_{t-1} + \beta_2 Z_{t-i} + v_2 t \quad \dots\dots (3)$$

Where Z represents the dependent variable shows the independent variable in the equation (2) and reversely in equation (3) and v_{1t} and v_{2t} are the error-correction term.

The overall F- and t-statistics are used to determine the presence of the long-run relationship. The null hypothesis is tested by using the generalized F-statistics. The test involves asymptotic critical –value bound, depending on whether the variables are integrated of order 0 or 1 [i.e.I(0) or I(1)]. Two sets of critical values are generated. One set refers to the I(1) series; the other refers to the I(0) series. The critical values for the I(1) are said to be the upper- bound critical values; the critical values for the I(0) series are referred to as lower- bound critical values (Narayan and Smyth,2005; Pesaran et al.,2001,2000; Pesaran and Shin,1999; Pesaran and Smith,1998;Pesaran and Pesaran,1997). To determine the order of integration of the series, we used the augmented Dickey-Fuller test (Dickey and Fuller,1981) unit root test.

If the computed F-statistics are above the upper bound, the null hypothesis of co-integration needs to be rejected, indicating evidence of a long-run equilibrium relationship between the variables, regardless of the order of the integration of the variables. If the test statistics fall below the lower bound, we cannot reject the null hypothesis of co-integration, indicating the absence of long run equilibrium relationships. If the test statistics falls between the bounds, a conclusive inference cannot be made without knowing the order of the integration of the underlying regressors. The order of lags in the ARDL model is selected by minimizing the Schwarz Bayesian criterion and Akaike information criterion.

According to Granger (1969), if variable Z forecast other variable X, that means Z Granger-causes X. If Z causes X and X does not causes Z, this is unidirectional causality. If Z does not cause X and X does not cause Z, both the variable are independent. For Granger analysis, autoregressive with lag length q (or p) estimate following model-

$$X_t = \gamma_1 + \sum_{i=1}^q a_{1i} X_{t-i} + \sum_{j=1}^q b_{1j} Z_{t-j} + \mu_{1t} \dots\dots\dots (4)$$

$$Z_t = \gamma_2 + \sum_{i=1}^q a_{2i} X_{t-i} + \sum_{j=1}^q b_{2j} Z_{t-j} + \mu_{2t} \dots\dots\dots (5)$$

Engle and Granger (1987) suggest, if two time series are co-integrated the ECM is preferable over VAR to interpret the dynamic changes in the short-run. For the purpose of capturing the long-run behavior error correction term ECT (which is lag value of residuals generated from the model estimate in level) is included in short-run equation. The ECM in first difference can be defined as-

$$\Delta Z_t = \gamma_2 + \sum_{i=1}^p a_{2i} \Delta X_{t-i} + \sum_{j=1}^p b_{2j} Z_{t-j} + ECT(-1) + \mu_{2t} \dots\dots\dots (6)$$

$$\Delta X_t = \gamma_1 + \sum_{i=1}^k a_{1i} \Delta X_{t-i} + \sum_{j=1}^k b_{1j} \Delta Z_{t-j} + ECT(-1) + \mu_{1t} \dots\dots\dots (7)$$

5. RESULTS AND DISCUSSION

The descriptive statistics for the variables interest rate and stock market indices (BSE Sensex) for empirical investigation purpose is shown in the Table I. Kurtosis and skewness values signify the symmetry in distribution. Skewness coefficient value greater than one (>1) is considered extreme (Chou, 1969) and low or high value of kurtosis reflect leptokurtic or platykurtic (Parkinson, 1987). Kurtosis and skewness values 3 and zero, indicate that the time series data is distributed normally. Table 1, that the variables mentioned above is not normal. Standard deviation value indicates that interest rate and BSE Sensex are less volatile.

Table 1: Descriptive Statistics of the Variables

Frequency of Data	Daily		Weekly		Monthly		Quarterly	
	LnBSE	IR	LnBSE	IR	LnBSE	IR	LnBSE	IR
Mean	9.964823	7.810758	7.824746	9.968365	9.968177	7.819199	7.819568	9.968562
Median	9.908651	7.990000	7.984000	9.907669	9.914857	7.952202	7.985317	9.908588
Maximum	10.29829	10.35000	10.27400	10.29210	10.27342	10.01150	9.146349	10.26035
Minimum	9.627410	0.250000	3.887500	9.652352	9.680361	5.343889	6.346029	9.709554
Std. Dev.	0.189273	0.932475	0.815852	0.189518	0.189973	0.774086	0.697506	0.191638
Skewness	0.224569	-1.41546	-0.134926	0.196466	0.208290	-0.036849	-0.323055	0.218710
Kurtosis	1.620670	11.87504	5.454724	1.594937	1.562528	4.585018	2.552051	1.507223
Jarque-Bera	111.0882	4581.284	66.32109	23.14851	5.599666	6.294282	0.515098	2.016432
Probability	0.000000	0.000000	0.000000	0.000009	0.060820	0.042975	0.772944	0.364869
Sum	12625.43	9896.230	2042.259	2601.743	598.0906	469.1519	156.3914	199.3712
Sum Sq. Dev.	45.35346	1100.799	173.0597	9.338417	2.129289	35.35331	9.243771	0.697781
Observations	1267		261		60		20	

For applying regression analysis on the time series data, stationary of time series variable is minimum requirement to minimize the problem of spurious regression that might arise if the series is non-stationary. Hence, time series is required to be made stationary first. Stationary of time series refers, variance and mean should be constant and co-variance value depends on the

distance of two time periods. The Augmented Dicky-fuller unit root test (Dickey and Fuller, 1979) is used to check stationarity of the time series (daily, weekly, monthly, quarterly) and test statistics is shown in Table 3. Interest rate series of daily, weekly, monthly and quarterly data is stationary at 1% or 5% level of significance. As the series is stationary at level, interest rate series is known as integrated of order zero, i.e I(0). BSE Sensex series of daily, weekly, monthly and quarterly data is stationary at 1% level of significance at their first difference. So, BSE Sensex series become is integrated of order one, i.e I(1).

Table 2: Result of ADF Unit Root Test

Type of Data	Variable	ADF Test Statistics	Comments	Order of integration
Daily	IR	-4.691251*	Rejected H0	I(0)
	D(IR)	13.867427*	Rejected H0	I(1)
	Ln BSE	-0.786097	Failed to reject H0	
	D(Ln BSE)	-33.24411*	Rejected H0	I(1)
Weekly	IR	-3.367555**	Rejected H0	I(0)
	D(IR)	-26.71713*	Rejected H0	I(1)
	Ln BSE	-0.907804	Failed to reject H0	
	D(Ln BSE)	-12.88028*	Rejected H0	I(1)
Monthly	IR	-2.765297***	Rejected H0	I(0)
	D(IR)	-5.758823*	Rejected H0	I(1)
	Ln BSE	-0.609072	Failed to reject H0	
	D(Ln BSE)	-7.980031*	Rejected H0	I(1)
Quarterly	IR	-2.721754***	Rejected H0	I(0)
	D(IR)	-4.597671*	Rejected H0	I(1)
	Ln BSE	-2.410317	Failed to reject H0	
	D(Ln BSE)	-5.578470*	Rejected H0	I(1)

For selecting optimal lag, traditionally VAR model is repeated by reducing the lag length value from large value to minimum. We select the optimal lag for which Akaike information criterion and Schwarz criterion have smallest value (Grasa, 1989; DeJong et al., 1992; Maddala and Gujarati, 2003). For analysis purpose optimal lag is selected on the basis of Haman-Quinn

information criterion (HQ), Final prediction error (FPE), Schwarz criterion (SC), Akaike information criterion (AIC), and Sequential modified LR test statistics that is suggested by Lutkepohl (1993). For the present study selected optimal lag length for daily data is 6, for weekly data is 2, for monthly data is 2 and for quarterly data is 3. As both the series are integrated of the order 1 and zero for daily, weekly, monthly and quarterly data series, so F-statistics is applied to check the existence of long-run co-integration among variables. ARDL bound test is preferable over Johansen co-integration test for co-integration analysis as Johansen co-integration test can be used when both the series are of same order one i.e. $I(1)$ and results are highly sensitive towards lag order selected.

Table- 2 Result of ARDL Bounds Test

Null Hypothesis: No long-run relationships exist

Type of Data	Test Statistic	Value	k	Dependent Variable
Daily	F-statistic	13.11485	1	Interest Rate
		1.525306	1	BSE indices
Weekly	F-statistic	7.394633	1	Interest Rate
		1.536923	1	BSE indices
Monthly	F-statistic	5.226624	1	Interest Rate
		3.065917	1	BSE indices
Quarterly	F-statistic	7.944100	1	Interest Rate
		13.95875	1	BSE indices

Critical Value Bounds

Significance	I0Bound	I1Bound
10%	4.04	4.78
5%	4.94	5.73
2.50%	5.77	6.68
1%	6.84	7.84

Table II indicate the results of ARDL bound test. The lower (I_0) and upper bound (I_1) critical values for F-statistics are taken from Narayan (2005). F-statistics value greater than lower and upper bound value confers the rejection of null hypothesis of no long-run relationship among the variables. There exists a uni-directional long-run relationship among both the variables, interest rate and BSE Sensex in case of daily data, when interest rate is used as dependent variable as the

value of F-statistics is greater than the lower (I_0) and upper bound (I_1), rejecting the null hypothesis of no co-integration. But no integration relation exist among both the variables when BSE Sensex is used as dependent variable as the F-statistics value is less than the critical values for the lower and upper bound accepting the null hypothesis and violating the bound testing criterion for the existence of co-integration relation in case of daily data.

In case of weekly data, null hypothesis of no long-run relationship is rejected as the value of F-statistics is greater than the critical values for lower (I_0) and upper bound (I_1), when interest rate is considered as a dependent variable. But the null hypothesis of no co-integration is accepted when BSE Sensex is considered as a dependent variable as the F-statistics value is less than the critical values for the lower and upper bound.

Null hypothesis of no long-run relationship is rejected in case of monthly data too, as the value of F-statistics is greater than the critical values for lower (I_0) and upper bound (I_1), when interest rate is considered as a dependent variable. But the null hypothesis of no co-integration is accepted when BSE Sensex is considered as a dependent variable as the F-statistics value is less than the critical values for the lower and upper bound.

But in case of Quarterly data, the null hypothesis of no long run relationship is rejected in both the cases that means interest rate and BSE Sensex are co-integrated in long-run when interest rate as dependent variable and BSE Sensex as a dependent variable respectively as the value of F-statistics is greater than the critical values for lower (I_0) and upper bound (I_1) in both the conditions.

Table- 3 Results of ARDL Co-integrating and Long Run Form for Daily Data

Dependent Variable: Interest Rate

Co-integrating Form

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(IR(-1))	-0.5442	0.031082	-17.508408	0.000
D(IR(-2))	-0.458316	0.031819	-14.404048	0.000
D(IR(-3))	-0.405533	0.031429	-12.903273	0.000
D(IR(-4))	-0.347388	0.029814	-11.651691	0.000
D(IR(-5))	-0.209962	0.026223	-8.006796	0.000
D(LnBSE)	-0.166627	0.081821	-2.036492	0.0419
ECT (-1)	-0.100064	0.019801	-5.053534	0.000

Long Run Coefficients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LnBSE	-1.665214	0.802981	-2.073791	0.0383
C	24.469366	8.005126	3.056712	0.0023
IR = -1.6652*LnBSE + 24.4694				

Wald-test statistics confirms that both the variables are co-integrated, so there exist a long run relationship among both the variables means interest rate affects BSE indices when daily data of both the variables is considered for analysis purpose. Negative value of the long - run coefficient indicates that interest rate is negatively correlated with stock market indices. If there is 1% increase (decrease) in stock market indices, interest rate will decrease (increase) by 1.67 %. That suggests that interest rate is negatively elastic towards change in stock market indices in long-run. In short-run also both the variable are negatively related, if there is 1% increase in stock market, interest rate will decrease by .167 %.

To capture the long run pattern between both the variables as both the variables are co-integrated, error correction term (ECT) (that is lag of the residuals generated through the model estimated at their levels) is included in the short-run equation. Error correction term's negative value shows the speed of adjustment in long-run and is significant at 5 % level. ECT absolute value less than one refers stable error correction mechanism as short-term dis-equilibrium converges to the long-term equilibrium level. The coefficient of ECM (-1) is equal to -0.100064 for short run model implying that the deviation from the long-run inequality is corrected by about 10 % each day. The error terms are homoskedastically normal and independent with no specific errors and LM-test confirms the absence of serial correlation.

Table- 4 Results of ARDL Co-integrating and Long Run Form for weekly data

Dependent Variable: Interest Rate				
Co-integrating Form				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(IR(-1))	-0.366434	0.051926	-7.056834	0.000
D(LnBSE)	-0.238613	0.118336	-2.016402	0.0448
ECT (-1)	-0.110824	0.029311	-3.780914	0.0002

Long Run Coefficients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LnBSE	-2.153089	1.07884	-1.995744	0.047
C	29.369531	10.762082	2.728982	0.0068
IR = -2.153089* LnBSE + 29.369531				

In case of weekly data, there exists a long-run relationship as both the variables are co-integrated. Negative value of the long-run coefficient confers that the interest rate is negatively elastic towards change in stock market indices. If the stock market indices will change by 1%, interest rate will negatively change by 2.15% in long-run. In short-run, also both the variables are negatively correlated. If there is 1% increase (decrease) in stock market indices, interest rate will decrease (increase) by .24% in short-run. Negative and significant value of error correction term (ECT) confirms the stability of the short-run equation and the deviation from the long-run inequality is corrected by about 11 % each day that is improved over daily data.

Table- 5 Results of ARDL Co-integrating and Long Run Form for monthly data

Dependent Variable: Interest Rate

Co-integrating Form				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(IR(-1))	0.227099	0.113576	1.999536	0.0506
D(LnBSE)	-0.500136	0.263693	-1.896665	0.0632
CointEq(-1)	-0.225519	0.071246	-3.165335	0.0025

Long Run Coefficients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LnBSE	-2.217717	1.230293	-1.802592	0.077
C	29.980014	12.272318	2.442897	0.0179
IR = -2.217717*LnBSE + 29.980014				

When monthly data of interest rate is regressed over stock market indices, there exists a long-run relationship among both the variables. Negative value of the long-run coefficient, -2.22 confers

that both the variables are negatively correlated in long-run. If there is 1% increase (decrease) in stock market the interest rate will decrease (increase) by 2.22%. In short-run also both the variables are negatively elastic towards each other. If there is 1% increase in stock market performance, interest rate will decrease by .50%, half of the changes in stock market performance. That means the changes are quite observable in case of monthly data. So, monthly data seems to be more appropriate for investment analysis purpose. Negative and significant value of error correction term (ECT) confirms the stability of the short-run relation. The coefficient of ECM (-1) is equal to -0.23 for short run model implying that the deviation from the long-run inequality is corrected by about 23 % each month. As we are moving from daily data to monthly data the speed of adjustment is improving.

Table- 6 Results of ARDL Co-integrating and Long Run Form for quarterly data

Dependent Variable: Interest Rate

Co-integrating Form				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LnBSE)	3.335607	2.809099	1.187429	0.2535
ECT (-1)	-0.808851	0.23304	-3.470872	0.0034
Long Run Coefficients				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LnBSE	-1.871639	0.815572	-2.29488	0.0366
C	26.4882	8.143227	3.252789	0.0054
IR = -1.871639 * LnBSE + 26.4882				

In case of quarterly data, there exist long run relationships between interest rate and stock market indices when interest rate is considered as a dependent variable. Negative value of the long - run coefficient indicates that both the variables are negatively correlated. If there is 1% increase (decrease) in the stock market indices the interest rate will decrease (increase) by 1.87 %. That suggests that interest rate market is negatively elastic towards stock market change in long-run. But, in short-run both the variable are positively correlated as the value of coefficient is positive. If there is 1% increase in the stock market performance, interest rate will also increase by 3.35%. Error correction term (ECT) value is -0.81, that is negative and significant that is the requirement for the stable short-run relationship and the absolute value less than one refers stable error correction mechanism as short-term dis-equilibrium converges to the long-term equilibrium

level. Deviation from the long-run inequality is corrected by about 81 % in each quarter. In case of quarterly data the deviations very quickly recover the equilibrium.

The scope for first or higher order auto-correlation is completely rejected. The error terms are homoskedastically normal and independent with no specific errors. For identifying the bilateral or unidirectional long-term/short-term causality among the Indian stock market, Pakistani stock market and Sri-Lankan stock market, Granger causality (1969) is applied and results are summarized in Table 7.

Table-7 Pair-wise Granger Causality Tests

	IR & LnBSE	Short-run causality results		Long-run causality results t- statistics (p- value) ECM (-1)	Decision of short-run causality	Decision of Long- run causality
		F-statistics (p-value)				
		IR	LnBSE			
Daily	IR		1.00865 (0.4180)	-0.100064 (0.0000)	No Causality	IR ---> LnBSE
	LnBSE	1.14501 (0.3338)		NO	No Causality	No Causality
Weekly	IR		2.52317 (0.0822)	-0.110824 (0.00024)	No Causality	IR ---> LnBSE
	LnBSE	1.94848 (0.1446)		NO	No Causality	No Causality
Monthly	IR		5.20823 (0.0086)	-0.225519 (0.0025)	IR ---> LnBSE	IR ---> LnBSE
	LnBSE	4.85435 (0.0116)		NO	LnBSE---> IR	No Causality
Quarterly	IR		3.74553 (0.0488)	-0.808851 (0.0034)	IR ---> LnBSE	IR ---> LnBSE
	LnBSE	0.98159 (0.4398)		NO	No Causality	No Causality

In case of daily data, the null hypothesis of no short-run causality is accepted as the value of the F-statistics are not significant. But in case of long-run, the null hypothesis of no causality is

rejected as the value of the error correction term is negative and significant, considering interest rate as a dependent variable. So, there exist a uni-directional long-run causality from interest rate to stock market indices i.e interest rate granger causes stock market indices. There is no long-run causality relationship exist among both the variable when stock market indices is considered as dependent variable.

Similar results occur in case of weekly data, as the null hypothesis no causality is accepted in case of short-run as the value of F-statistics is not significant. Uni-direction long-run causality relationship from interest rate to stock market indices exist. That means interest rate granger causes stock market indices in long-run.

The null hypothesis of no short-run causality is rejected as the value F-statistics and p-value are significant at 1% and 5%. So, in short-run there is bi-directional causality exist. Interest rate granger causes stock market and stock market granger causes interest rate. But in long-run, there exist a uni-directional long-run relationship exist as the value of the error correction term is negative and p-value is significant at 1% considering interest rate as a dependent variable. So, in long-run, interest rate granger causes stock market.

In case of quarterly data, there exist a uni-directional short-run causality from interest rate to stock market exist as F-statistics is significant and results are supported by p-value at 5% significance level rejecting the null hypothesis of no short-run causality. So, in short-run, interest rate granger causes stock market. In long-run also uni-directional causality exist as the error correction term is negative, less than one and p-value is significant at 1% significance. So, in long-run interest rate granger causes stock market.

6. Conclusion and Discussion

In this paper effort is directed to resolve the issues of variation in the finding of different studies conducted on stock market and interest rate relation across globe. This variation might arise because of the nature of the data (daily, weekly, monthly, quarterly) used in different studies. To explore this issue same daily data of both the variable is converted in to weekly, monthly and quarterly and analyses to extract and verify the relation between both the variables. Results accept the null hypothesis of co-integration relation between both the variables as there exist a long-run relationship between the variables when interest rate is regressed over stock market but no relation exist when stock market is regressed over interest rate expect for quarterly data where

bi-directional long-run relationship exist. Analysis support the null hypothesis of negative relation between both the variables for daily, weekly, monthly and quarterly data during long-run and short-run except quarterly data where a short-run positive relation exist between the variables supporting the finding of Ratanapakorn and Sharma (2007). The outcome of the study contradicts the research finding of Liu and Shrestha (2008) of no relation on Chinese stock market. Results of the study support that if the cost of borrowing of the fund increases, it will adversely influence the performance of the stock market and follow down-turn in long-run as well as in short-run. So, the investors are required to consider interest rate changes before taking any investment decision. Results also confirm that the changes in interest rate are directed and controlled by the stock market movement. Market performance provides basis for regulating changes in interest rate. Investors are required to keep the stocks for longer period of time as the speed of the adjustment is in increasing order from daily data to quarterly data and more than 80% in case of quarterly data. Market will adjust and accommodate all kind of interest rate shock and achieve its equilibrium very quickly within a quarter. So, equity investors are required to keep their investments at least for a quarter to get satisfactory return.

The results of granger causality analysis confirm uni-directional long-run causality from interest rate to stock market supporting the null hypothesis of the study and supporting the outcome of the study by Panda (2008) on Indian stock market. There is no short-run casualty relationship exist in case of daily data and weekly data. Interest rate granger causes stock market and stock market granger causes interest rate, a bi-directional short-run casualty relation exists in case of monthly data and uni-directional short-run causality from interest rate to stock market exist in case of quarterly data.

Overall the relation between interest rate and stock market is negative in long-run and short-run confirming that this negative relation is independent of the nature of the data.

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